

The two-good quasilinear economy

There are J firms, I consumers, and $L = 2$ goods. Good 1 is a consumption good and an output for each firm. Good 2 is both a consumption good and an input for each firm. The relative price of good 1 in terms of good 2 is $p \geq 0$. The price of good 2 is normalized to 1. I will assume that this is a *private ownership economy*: each firm is owned entirely by consumers.

FIRMS

A production plan for firm j is a point $y_j = (q_j, -z_j)$, where q_j is output of good 1 and z_j is the input of good 2. Firm j 's production function is a C^1 function f_j with $f_j' > 0$ and $f_j(0) = 0$. Its production set is $Y_j = \{(q_j, -z_j) \mid 0 \leq q_j \leq f_j(z_j), z_j \geq 0\}$ and its cost function is $c_j = f_j^{-1}$. Firm j 's problem is to choose $y_j \in Y_j$ to maximize $pq_j - z_j$. Since any solution to this problem is technologically efficient, we may set $z_j = c_j(q_j)$ and write firm j 's profit-maximization problem as ($p \geq 0$)

$$(\mathbf{PM}_j) \quad \pi_j(p) = \max_{q_j \geq 0} \{pq_j - c_j(q_j)\}$$

with firm j 's input use equal to $z_j = c_j(q_j)$. I record for later use the first-order conditions:

$$p - c_j'(q_j) \leq 0, \quad q_j[p - c_j'(q_j)] = 0, \quad q_j \geq 0.$$

Denote the firm's supply of good 1 at p by $s_j(p)$. [Note: if in addition c_j is strictly convex with $c_j'(\infty) = \infty$, then $s_j(p)$ is nonempty, single-valued, and continuous at any $p \geq 0$.]

CONSUMERS

A consumption plan for consumer i is a point (x_i, m_i) , where x_i is i 's consumption of good 1 and m_i is i 's consumption of good 2. Consumer i 's consumption set is $X = \mathbb{R}_+ \times \mathbb{R}$. It is endowed with $\omega_i > 0$ units of good 2, and ownership shares $(\theta_{i1}, \dots, \theta_{iJ})$, where each $\theta_{ij} \geq 0$ and $\sum_i \theta_{ij} = 1$ for $j = 1, \dots, J$. Consumer i 's price-dependent wealth, $w_i(p)$, is

$$w_i(p) = \omega_i + \sum_j \theta_{ij} \pi_j(p).$$

The budget constraint is, for $(x_i, m_i) \in X$,

$$(\mathbf{B}_i) \quad px_i + m_i \leq w_i(p).$$

Consumer i 's preferences are quasilinear in good 2 and represented by $u_i(x_i, m_i) = \phi_i(x_i) + m_i$ on X , where ϕ_i is C^1 with $\phi_i' > 0$. Consumer i chooses $(x_i, m_i) \in X$ satisfying (B_i) to maximize $u_i(x_i, m_i)$. Since u_i is strictly increasing, (B_i) holds as an equality at any solution, so we can write consumer i 's utility-maximization problem as

$$(\mathbf{UM}_i) \quad \max_{x_i \geq 0} \{\phi_i(x_i) - px_i\},$$

with consumer i 's consumption of good 2 equal to $m_i = w_i(p) - px_i$. I record the first-order conditions for future use:

$$\phi_i'(x_i) - p \leq 0, \quad x_i[\phi_i'(x_i) - p] = 0, \quad x_i \geq 0.$$

Denote consumer i 's demand for good 1 at p by $d_i(p)$. [Note: if in addition ϕ_i is strictly concave with $\phi_i'(\infty) = 0$, then $d_i(p)$ is nonempty, single-valued, and continuous at any $p > 0$.]

An *allocation* for this economy is a point $\xi = ((x_1, m_1), \dots, (x_I, m_I), y_1, \dots, y_J) \in X^I \times \prod_{j=1}^J Y_j = \mathbb{A}$. Let $D(p) = \sum_i d_i(p)$ be the aggregate demand for good 1, $S(p) = \sum_j s_j(p)$ the aggregate supply of good 1, and $Z(p) = D(p) - S(p)$, the *excess demand* for good 1 at p . The definition of a *competitive equilibrium* specializes MWG's Definition 10.B.3 to a two-good quasilinear economy.

WELFARE IN THE 2-GOOD QUASILINEAR MODEL (OR: CONCAVE PROGRAMMING AND THE INVISIBLE HAND). Consider the problem (maximize the sum of utilities over all *feasible* allocations—defined by the two constraints)

$$\begin{aligned}
 (\mathbf{WM-0}) \quad & \max_{\xi \in \mathbb{A}} \sum (\phi_i(x_i) + m_i) \\
 \text{s.t.} \quad & \sum x_i \leq \sum q_j \\
 & \sum m_i + \sum z_j \leq \sum \omega_i.
 \end{aligned}$$

By the Lemma in the *Pareto Optimality* notes, any solution to **(WM-0)** is Pareto Optimal. Let each ϕ_i and f_j be **concave** as well as C^1 with positive derivatives. Then: *If (ξ^*, p^*) is a Competitive Equilibrium, then ξ^* solves problem **(WM-0)**.* Since any solution to (PM_j) is technically efficient the input allocation $(z_1, \dots, z_J)$ equals $(c_1(q_1), \dots, c_J(q_J))$, where for $j = 1, \dots, J$, $c_j(q_j) = f_j^{-1}(q_j)$. And since the objective is increasing in m_i for every i , the second constraint holds as an equality (and since we are ignoring the boundary condition on the second good) use the constraint to substitute $\sum m_i$ out of the objective to reformulate the problem as

$$\begin{aligned}
 (\mathbf{WM}) \quad & \max \{ \sum \phi_i(x_i) + \sum \omega_i - \sum c_j(q_j) \} \\
 \text{s.t.} \quad & \sum x_i \leq \sum q_j,
 \end{aligned}$$

where the maximum is taken over all $\xi_1 = (x_1, \dots, x_I, q_1, \dots, q_J) \geq 0$ (an application of the Principle of Optimality).

I now argue that the good-1 component of ξ^* solves **(WM)**: since the objective function is concave and the constraint functions linear, any solution to the first-order conditions solves the problem; *and the first-order conditions hold at ξ^* with constraint multiplier equal to $p^* \geq 0$.* Since ξ^* solves **(WM-0)** it follows that ξ^* is *Pareto Optimal*: if any consumer i strictly prefers another feasible (i.e. satisfying the constraints of **(WM-0)**) allocation ξ to ξ^* , then some other consumer i' strictly prefers ξ^* to ξ .

Next, we'll together show that $(q_1^*, \dots, q_J^*)$ solves the following aggregate cost minimization problem at $Q = \sum q_j^*$,

$$\begin{aligned}
 (\mathbf{ACM}) \quad & C(Q) = \min_{(q_1, \dots, q_J) \geq 0} \sum c_j(q_j) \\
 \text{s.t.} \quad & \sum q_j \geq Q;
 \end{aligned}$$

and that $(x_1^*, \dots, x_I^*)$ solves the following aggregate utility maximization problem at $X = \sum x_i^*$. (MWG, 10.C.3 & 10.C.4.)

$$\begin{aligned}
 (\mathbf{AUM}) \quad & \Phi(X) = \max_{(x_1, \dots, x_I) \geq 0} \sum \phi_i(x_i) \\
 \text{s.t.} \quad & \sum x_i \leq X.
 \end{aligned}$$

By the Principle of Optimality, the Pareto Optimal aggregate consumption of good 1 maximizes $W(Q) = \Phi(Q) - C(Q)$ subject to $Q \geq 0$.

The economy has a *representative consumer* with preferences represented by $U(X, M) = \Phi(X) + M$. The economy also has a *representative firm* with cost function given by $C(Q)$, which describes the economy's aggregate production set. The representative-consumer fact follows since every consumer's preferences are quasilinear wrt good 2, and so the economy's family of indirect utilities takes the Gorman form; by contrast, the representative-firm fact is general, and does not depend on what the firms' technologies are; see MWG Proposition 5.E.1 (on ASMT Va, # 8.)

EXISTENCE

The next lemma is an immediate consequence of the Intermediate Value Theorem.

Lemma (Existence). *If excess demand $Z(p)$ is nonempty, single-valued, and continuous at every $p > 0$; and there are two positive prices p_0 and p_1 with $Z(p_0) \geq 0 \geq Z(p_1)$, then there is a $p^* > 0$ such that $Z(p^*) = 0$; that is, a competitive equilibrium exists.*

As for the conditions that Z is nonempty, single-valued, and continuous, suppose that each ϕ_i and f_j is (i) strictly concave and (ii) C^1 with (iii) $\phi'_i > 0$, $f'_j > 0$, (iv) $\phi'_i(\infty) = 0 = f'_j(\infty)$, and (v) $f_j(0) = 0$. Then each d_i and s_j are nonempty, single-valued, and continuous at every $p > 0$. Why? First the K-T conditions are both necessary and sufficient for each agent's problem. Second, each agent's problem has a solution for each $p > 0$: for consumer i , either $\phi'_i(x) = p$ has a positive solution (in which case $d_i(p) = (\phi'_i)^{-1}(p)$) or it doesn't (in which case $d_i(p) = 0$). This follows from continuity of ϕ'_i and the Inada condition. So the demand for i is $\max\{0, (\phi'_i)^{-1}(p)\}$. Since the maximum of continuous functions is continuous, i 's demand is continuous. The same argument works for firms. And since the sum of continuous functions is continuous, $Z(\cdot)$ is continuous.

Since $\lim_{p \rightarrow 0} Z(p) = \infty$ (by strong monotonicity of preferences), there is a $p_0 > 0$ with $Z(p_0) > 0$. Set $p_1 = \max\{\phi'_1(0), \dots, \phi'_I(0)\}$. Then $Z(p_1) \leq 0$ (since $D(p_1) = 0$ and $S(p_1) \geq 0$). Under conditions (i)-(v), a competitive equilibrium exists.

Nonconvexities can lead to nonexistence. As you proved on ASMT IV, the (PM) problem might not have a solution if returns to scale are nondecreasing. This possibility is the most important *economic* reason why a competitive equilibrium might not exist. The following fact is an easy corollary of what you've already proved.

Fact (Increasing Returns and Nonexistence). *If for some firm j , $f_j(tz) > tf_j(z)$ for every $z > 0$ and $t > 1$ (and $f_j(0) = 0$) then there is no competitive equilibrium with $q_j > 0$.*

Corollary: If the *increasing returns* condition holds for *every* firm and good 1 is *essential* for some consumer i (see my notes on Essentiality), then there is no competitive equilibrium.

Nonconvex preferences can also lead to nonexistence. A problem is that the aggregate demand correspondence need not be convex. Happily in this case, this nonexistence is an artifact of using a mathematical model with a finite number of consumers to model a situation in which each consumer is supposed to be *negligible* (have no effect on prices). Aumman (1964) suggested that an appropriate model of a competitive economy is one with a *continuum* of consumers. He used this idea in 1966 to prove that, with a *continuum* of consumers, say $I = [0, 1]$, then the *aggregate* demand correspondence will be convex-valued, even though the individual correspondences need not be.

THE FIRST WELFARE THEOREM WITH NONCONVEXITIES

Continue to impose all the assumptions of the two-good quasilinear economy except the concavity assumptions on the ϕ_i and f_j functions; in particular preferences and production sets need not be convex. Because of quasilinearity, there is a (positive) representative consumer with preferences represented by $U(X, M) = \Phi(X) + M$, where $\Phi(\cdot)$ is the value function in problem (H).

I will show that any Competitive Equilibrium allocation is Pareto Optimal. To begin suppose that ξ^* is a feasible allocation that is not Pareto Optimal. I will show that ξ^* cannot be a Competitive Equilibrium allocation. If there is *no* price p^* such that each consumer i demands the consumer- i component of ξ^* , then we are done. So suppose there is a price p^* such that each consumer i demands the consumer- i component of ξ^* . Let Q^* be the aggregate demand for good 1 at p^* , which must solve the representative consumer's problem, choosing $Q \geq 0$ to maximize $\Phi(Q) - p^*Q$. I will argue that i) Q^* does not maximize $\Phi(Q) - C(Q)$; and ii) Q^* does not maximize aggregate profit (so cannot solve some firm's problem PM_j).

Regarding i), the consumer allocation maximizes (H) at Q^* (since x_i^* maximizes $\phi_i(x_i) - p^*x_i$ the allocation $(x_1^*, \dots, x_I^*)$ must maximize $\sum \phi_i(x_i) - p^*Q^*$). If ξ^* is not Pareto Optimal, then there is some other output $Q' \geq 0$ and allocation of the stock $(Q', \omega - C(Q'))$ that every consumer weakly prefers and some consumer strictly prefers. That fact in turn implies i): $\Phi(Q') - C(Q') > \Phi(Q^*) - C(Q^*)$ or equivalently, $[\Phi(Q') - p^*Q'] + [p^*Q' - C(Q')] > [\Phi(Q^*) - p^*Q^*] + [p^*Q^* - C(Q^*)]$. Since Q^* solves the representative consumer's UM problem at p^* , the last inequality implies ii) that Q^* does *not* maximize aggregate profit at p^* .